Problem Statement & Objectives

Problem Statement

In manufacturing and production management environments, large volumes of data are recorded related to production processes, including Production, Batches, Products, and Downtime events.

However, the main problem is that this data is often not effectively utilized in an analytical way, which leads to:
Lack of clarity regarding the causes of production stoppages
Difficulty in measuring the efficiency of each batch
Weak ability to make fast, data-driven decisions
Absence of a clear understanding of the relationship between production and downtime

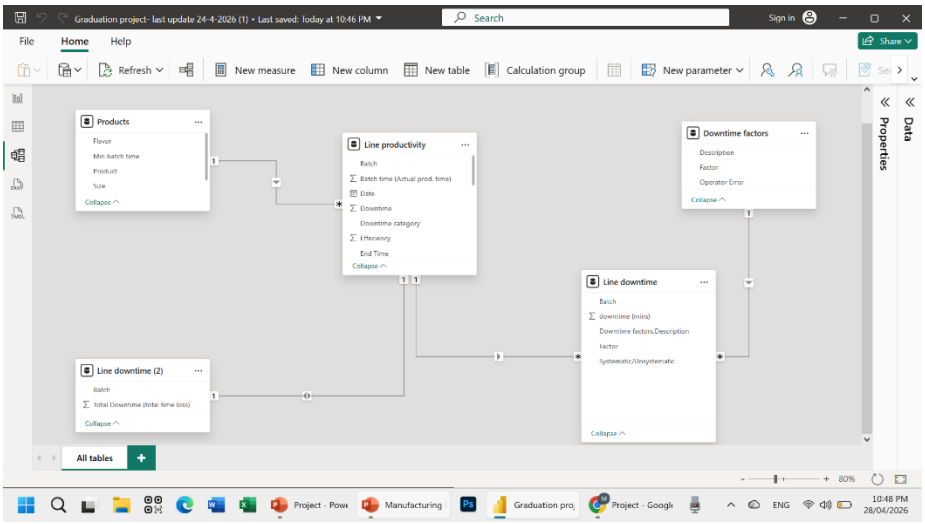
Therefore, there is a strong need for an analytical system that transforms this raw data into meaningful insights, enabling management to improve performance and reduce losses.

Objectives

This project aims to:

- Perform detailed analysis of production data
- Study the relationship between production and downtime impact
- Analyze the performance of each batch independently
- Identify the most efficient products and those causing delays
- Calculate key performance indicators (KPIs) such as:
- Production Efficiency
- Downtime Rate
- Batch Performance
- Provide an interactive dashboard that supports faster and more accurate decision-making for management

Database Design & Data Modeling



Data Flow & System Behavior

Data Flow (DFD Overview)

Data is generated from production machines and captured by sensors that monitor machine status (Running, Idle, or Breakdown).
This data is sent to the system where downtime events are detected and classified.

All records are stored in a central database and later used to calculate key performance indicators (KPIs) such as:
Utilization Rate
Idle Time
Production Efficiency

System Sequence

- Machine sends real-time status data
- Sensor collects and transmits data to the system
- System detects downtime events
- Events are stored in the database
- KPIs are calculated and updated
- Dashboard displays insights to the production manager

UI/UX Design & Prototyping

The system interface is designed to provide clear and real-time visibility of machine performance and downtime events. The main goal is to help production managers quickly understand system status and make decisions efficiently.

Wireframes & Mockups
The system includes simple dashboards showing:

- Machine status (Running / Idle / Breakdown)
- Downtime summary
- KPIs such as utilization and efficiency
- Production reports by batch and time period

UI/UX Guidelines
Design principle: Simple, clear, and data-focused interface
Color scheme: Green (Running), Yellow (Idle), Red (Downtime)
Typography: Clean and readable fonts for industrial dashboards
Accessibility: High contrast visuals for easy reading in factory environments

System Deployment & Integration

Technology Stack
Frontend: React / HTML / CSS (Dashboard interface)
Backend: Node.js / Python (Data processing & logic)
Database: MySQL / PostgreSQL (Production and downtime data storage)

Deployment Diagram (Concept)

The system is deployed in a way where:

- Sensors and machines send data from the factory floor
- Backend server processes and analyzes the data
- Database stores all production records
- Dashboard is accessible to managers via web interface

Component Diagram

The system consists of:

- Data Collection Layer (Sensors & Machines)
- Processing Layer (Backend System)
- Storage Layer (Database)
- Presentation Layer (Dashboard/UI)

Additional Deliverables

Testing & Validation

The system is validated through:

Unit Testing: Testing each module individually
Integration Testing: Ensuring system components work together
User Acceptance Testing (UAT): Verifying dashboard usability for production managers

Deployment Strategy

The system is hosted on a cloud or local server
Data is continuously streamed from machines to backend
Scalable architecture allows handling multiple production lines
Future expansion supports real-time analytics and alerts